



BOILEAU GUILLAUME

System Performance Analysis Engineer | Modelling, Statistics, Diagnostics & Complex Systems

@ guillaume.boileaupro@gmail.com
Guillaume-Boileau-2

+33 6 59 94 85 84
0000-0002-3576-6968

Alpes-Maritimes, FRANCE

guillaume-boileau-phd-891b81265

EXPERIENCE

Software Engineer – System Integration, Operational Performance & Production Diagnostics

VEV Platform Services France

2025 – 2026

Valbonne, France

- Developed and maintained web and backend services for an EV charging CPMS platform using **TypeScript, Angular and Node.js**.
- Contributed to software integration activities involving operational charging infrastructure, hardware interfaces and field equipment constraints.
- Analysed software behaviours, operational data exchanges, communication issues and service anomalies affecting production continuity.
- Investigated data inconsistencies, abnormal behaviours and system-level interactions between software services and connected charging equipment.
- Supported monitoring, verification, correction and follow-up of operational data flows, including FTP-based exchanges.
- Contributed to issue diagnosis, prioritization and technical follow-up with stakeholders in an industrial operational context.
- Strengthened experience with production constraints, maintainability, service robustness and software systems connected to physical infrastructure.

Senior R&D Engineer – System Performance, Modelling, Validation & Technical Coordination

Laboratoire Lagrange, Observatoire de la Côte d'Azur, CNES

2023 – 2025

Nice, France

- Contributed to engineering and analysis activities for the **LISA ESA/NASA space mission**, involving complex system-level performance constraints, scientific requirements and high modelling rigor.
- Developed **CATpyLISA**, a Python platform for large-scale model integration, comparison, analysis and validation workflows.
- Conducted deep technical studies on complex system behaviours, weak-signal environments, numerical models, uncertainty sources and performance limitations.
- Analysed sensitivity to modelling assumptions, numerical parameters, data-processing choices, configuration changes and validation conditions.
- Identified bottlenecks, inconsistencies, deviations, critical behaviours and limiting factors in complex software and analysis chains.
- Structured validation workflows to compare outputs, verify expected behaviours, detect deviations and consolidate technical evidence.
- Supported activities aligned with **V-cycle engineering logic**: requirements interpretation, validation planning, consistency checks, result verification and anomaly analysis.
- Produced technical analyses supporting model selection, risk identification, performance interpretation and decision-making.
- Maintained traceability of configurations, assumptions, parameters, validation results and technical conclusions.
- Coordinated technical activities involving contributors, research engineers, technicians and Master's thesis interns.
- Operated in a collaborative international framework requiring technical English, structured reporting, versioned workflows and rigorous documentation.

Data Scientist – Instrument Performance, Noise Modelling & Statistical Diagnostics

Universiteit Antwerpen, Belgium

2021 – 2023

Antwerp, Belgium

- Analysed instrumental and environmental effects impacting the sensitivity, stability and performance of precision measurement systems.
- Developed Python-based analysis pipelines for noise source identification, uncertainty estimation, Bayesian inference and statistical modelling.
- Conducted deep technical studies on degradation patterns, dominant contributors, system limitations and correlations between environmental conditions and measurement performance.
- Investigated unexpected behaviours, weak points and possible mitigation directions using statistical diagnostics and reproducible workflows.
- Built traceable analysis workflows using Unix environments, scripting, source control practices and structured documentation.
- Contributed to studies connected to gravitational-wave detector systems, where performance, noise control, robustness and system-level sensitivity are critical.
- Communicated complex technical results to scientific and engineering stakeholders through reporting, presentations and collaborative review.

SUMMARY AND STRENGTHS

System performance and technical analysis engineer with strong experience in modelling, statistical diagnostics, simulation workflows, software/data validation and complex system analysis. Background developed in international space and precision-instrumentation environments, complemented by recent industrial software experience involving operational data flows, hardware/software interfaces and production diagnostics.

Experienced in **deep technical studies, performance analysis, statistical modelling, uncertainty analysis, bottleneck identification, anomaly investigation, Python automation, Unix/Linux environments, scripting, source control, technical documentation and cross-functional coordination**. Able to analyse complex systems, identify limiting factors, consolidate technical evidence, communicate insights clearly and support improvement directions in high-constraint environments. Relevant for **system performance analysis, software/hardware interaction diagnostics, advanced modelling, statistical analysis, simulation workflows, technical studies, validation, Unix-based engineering and collaborative multi-site environments**.

FIT FOR ARM SYSTEM PERFORMANCE

- Strong experience conducting deep technical studies on complex systems, performance limitations, uncertainty sources and degraded behaviours.
- Solid background in statistics, Bayesian inference, signal processing, modelling, simulation and sensitivity analysis.
- Experience identifying bottlenecks, deviations, numerical behaviours, weak points and possible improvement or mitigation directions.
- Hands-on experience with **Python, C/C++ foundations, Unix/Linux**, Bash scripting and Git/GitLab.
- Used to analysing system-level behaviours at the interface between software, data, modelling assumptions and operational constraints.
- Comfortable producing structured technical evidence, documenting assumptions and communicating results to multidisciplinary teams.
- Familiar with high-performance and high-constraint technical contexts through space mission analysis, precision instrumentation and detector performance studies.
- Motivated to extend existing system-performance expertise toward CPU, interconnect and cloud/server SoC environments.

TEAM MANAGEMENT

- Supervision of **3 Master's thesis interns (M2)** during engineering, modelling and software-development phases.
- Coordination of technical activities involving **research engineers and technicians on a data processing and validation pipeline** for the **LISA space mission**.
- Experience in organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●

Junior R&D Engineer – Signal Analysis, Simulation & System Performance Assessment

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 Nice, France

- Conducted research and engineering-oriented analysis for the preparation of gravitational-wave space missions, with a strong focus on LISA.
- Developed methods for the separation of weak overlapping signals in complex low-SNR datasets.
- Built simulation and analysis workflows to evaluate system performance under realistic noise, foreground and uncertainty conditions.
- Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- Investigated noise sources, correlations and sensitivity limits through diagnostic analyses relevant to precision instruments.
- Produced peer-reviewed results contributing to the understanding of performance limits for future gravitational-wave space missions.
- Developed strong expertise in modelling, uncertainty analysis, signal processing, technical validation and scientific software workflows.

PROFESSIONAL TRAINING

Supervised Machine Learning

DeepLearning.AI – Andrew Ng

2020 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

2025 France Université Numérique

- Training focused on reproducible research practices, computational documentation, data management, version control and transparent analysis workflows.
- Practical exposure to Git/GitLab, Jupyter notebooks, structured documentation, data handling and reproducible software environments.

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026 On line

- Completed courses covering Python, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, SQL, Machine Learning, AI, Linux, C/C++, web development and software engineering fundamentals.
- Strengthened practical skills in full-stack development, data analysis, algorithmic thinking, database querying, Linux administration and modern software tooling.

PROJECTS

LISA Consortium – Software, Modelling & System Performance Analysis

ESA/NASA Space Mission – Large-scale International Collaboration

2018 – now

Contribution to software, analysis and validation workflows for the LISA space mission within an international engineering and scientific collaboration involving ESA, NASA and multiple research institutes.

Role: Contribution to software workflows, technical coordination, complex signal-processing activities, model integration, validation campaigns, performance assessment and collaborative engineering tasks.

System performance relevance: Work involving simulation, model comparison, weak-signal analysis, sensitivity studies, uncertainty analysis, bottleneck identification, validation evidence and performance limitation assessment.

Project relevance: Work at the interface between software development, scientific requirements, data processing, validation workflows, technical contributors and international coordination.

Program scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Einstein Telescope and LIGO–Virgo–KAGRA Collaborations

International Scientific and Engineering Collaborations

2021 – now

Contribution to collaborative developments in data analysis, signal characterization, software methods and technical investigations within the LIGO–Virgo–KAGRA and Einstein Telescope collaborations.

Role: Development of analysis tools, contribution to technical activities and support to complex software workflows in high-standard collaborative environments.

System impact: Contributed to studies on environmental and instrumental noise sources, supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

Environment: Large-scale international collaborations involving strong technical requirements, structured methods, technical review, versioned software workflows and coordinated contributions.

STRENGTHS

System performance & technical analysis

System Performance Analysis

Software Performance Analysis

Hardware Performance Analysis

Performance Modelling

Bottleneck Identification

Deep Technical Studies

Workload Characterisation

Statistical Modelling

Bayesian Inference

Simulation Workflows

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 Nice, France

Selective Graduate Program in Fundamental Physics

Université Paris-Saclay

2016 – 2018 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 Orsay, France

Uncertainty Analysis Sensitivity Analysis Signal Processing Noise Modelling Technical Diagnostics Anomaly Analysis

Systems & architecture

Systems Engineering Complex Systems System Architecture Computer Architecture Awareness CPU Performance Interest Interconnect Performance Interest
Pre-Silicon Environment Interest Software / Hardware Interfaces
Mission-Level Performance Validation Workflows Integration Testing Root Cause Analysis Configuration Follow-Up Technical Evidence

Technical skills

Python C / C++ Unix / Linux Bash Shell Scripting Git / GitLab Source Control Docker SQL Pandas Matlab TypeScript Node.js Jira
Confluence OpenSearch

Engineering environment

Reproducible Workflows Structured Documentation Technical Reporting Multi-Site Collaboration Cross-Functional Coordination Technical English Compiler Awareness
Device Driver Awareness Cloud / Server Interest Arm Architecture Interest Learning Agility

Applications: system performance analysis, software/hardware performance diagnostics, modelling, statistics, Unix-based engineering and complex system validation

Applications: system performance analysis, modelling, statistics, Unix-based engineering, software/hardware diagnostics and complex system validation